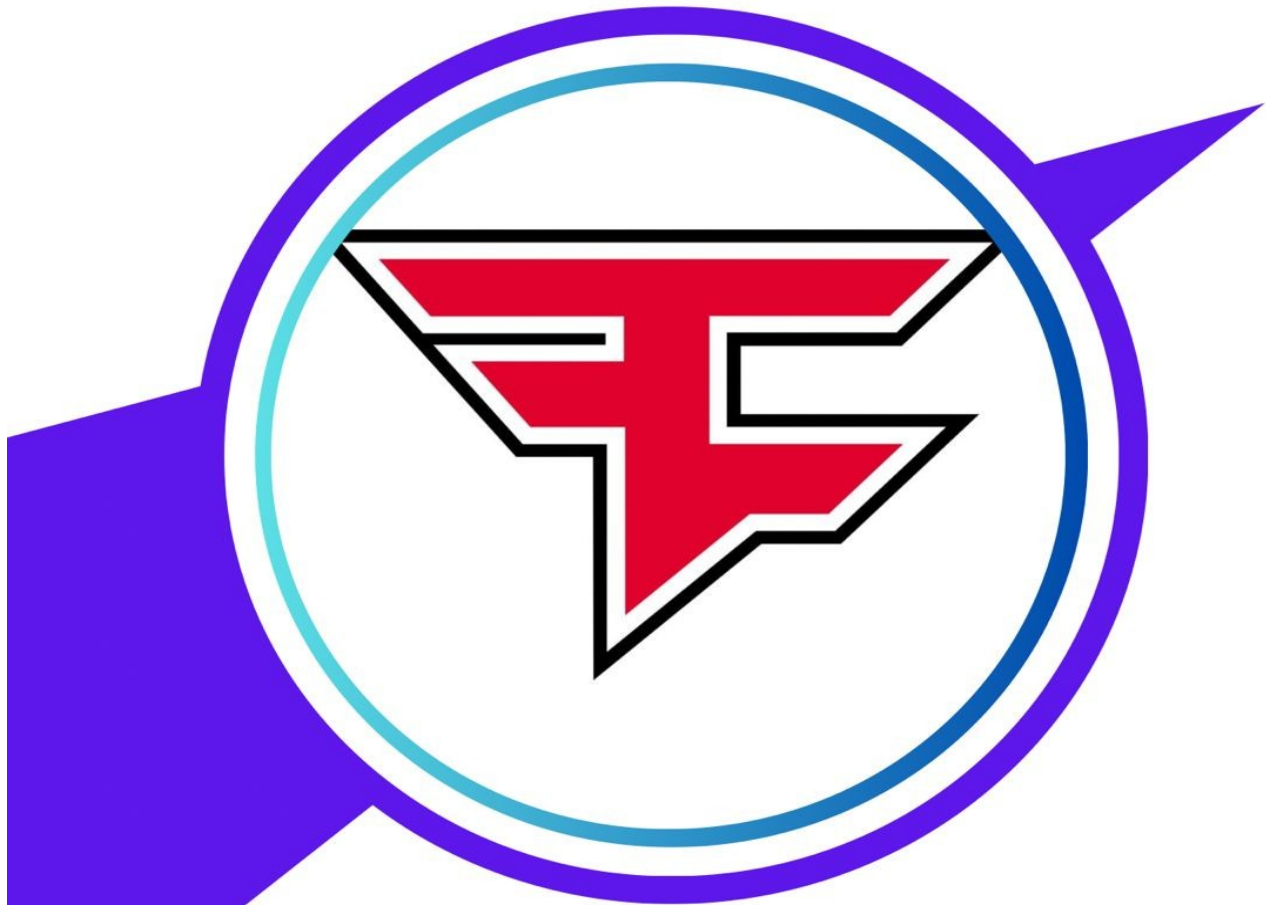




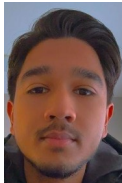
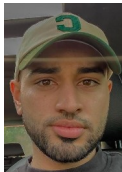
FAZE CLAN

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Project Phase 1 – Database Initial Study

2. Analyse the Company Situation

Background Information

Faze Clan Movements operates in the logistics and freight transportation industry. The company provides delivery services by transporting goods from suppliers to warehouses and retail outlets located in various regions. To support these operations, the organization maintains a fleet of trucks responsible for collecting and delivering cargo for different businesses.

Currently, the company manages much of its operational information using manual records and separate spreadsheet files. Important data related to shipments, drivers, vehicles, and customers is stored in multiple locations, which often leads to inefficiencies. This approach makes it difficult for staff to quickly retrieve accurate information and may result in duplicated data or missing records.

To improve the reliability and organization of its operations, the company plans to implement a centralized database system. This system will allow all operational information to be stored in one structured platform. The database will manage data related to deliveries, vehicles, drivers, routes, warehouses, cargo items, invoices, and payment records. By introducing this digital system, the company expects to improve accuracy, efficiency, and operational transparency.

2.1 Company Objectives

Faze Clan Movements aims to improve its logistics operations and service delivery by implementing a modern operational database system. The company has the following objectives:

- **Improve Delivery Efficiency**

The organization aims to improve delivery performance by organizing shipments more effectively and ensuring that trucks and drivers are assigned efficiently.

- **Centralized Data Management**

A central database will allow the company to store all operational data in one place. This includes information about customers, shipments, vehicles, drivers, and warehouse activities.

- **Improve Customer Satisfaction**

By implementing a digital system, the company will be able to provide customers with more accurate updates regarding delivery times, shipment progress, and order confirmations.

- **Improve Fleet Management**

The system will help the company monitor vehicle usage, maintenance schedules, inspections, and fuel consumption, ensuring that the fleet operates safely and efficiently.

- **Improve Financial Management**

The database system will assist the finance department with generating invoices, monitoring payments, and keeping accurate financial records related to transportation services.

- **Improve Reporting and Decision Making**

Management needs detailed reports on the following: deliveries, driver performance, vehicle usage, operational costs, as well as customer activity.

2.2 Company Operations

Faze Clan Movements operates in the logistics and freight transportation industry. The company coordinates transportation services between businesses that need goods delivered to various locations.

The company's operations include the following key activities:

Shipment Requests and Scheduling

Customers submit requests when they need goods transported from a pickup location to a destination. Dispatchers review these requests and organize delivery schedules based on vehicle availability and driver assignments.

Dispatch and Route Planning

The dispatch department assigns trucks and drivers to shipments. Efficient delivery routes are planned to reduce travel time and minimize fuel costs.

Cargo Transportation

During the delivery process, shipment status updates are recorded to indicate whether the cargo has been collected, is currently in transit, or has reached its destination.

Vehicle and Fleet Management

The company manages a fleet of vehicles that require regular maintenance and safety inspections. Monitoring fuel usage and maintenance schedules helps ensure reliable operations.

Warehouse and Depot Coordination

Some goods may be temporarily stored at warehouses or depots before final delivery. Warehouse staff manage cargo handling, storage, and preparation for dispatch.

Customer Service and Communication

Customer service representatives communicate with clients regarding shipment status, delivery updates, and service inquiries.

Billing and Financial Processing

Once shipments are completed, invoices are generated and sent to customers. The finance department records payments and manages financial transactions.

2.3 Company Organizational Structure

The organizational structure of Faze Clan Movements consists of several departments that work together to ensure efficient logistics operations.

Executive Management

Chief Executive Officer (CEO)

Oversees the strategic direction of the company.

Operations Director

Ensures that transportation activities run efficiently.

Operations and Dispatch Department

Dispatch Manager

supervises shipment scheduling and the allocation of drivers and trucks.

Dispatch Officers

Manages booking requests, coordinate deliveries, and communicate with drivers and customers.

Fleet Management Department

Fleet Manager

Monitors vehicle performance, maintenance schedules, and inspections.

Maintenance Technicians

Carries out repairs and ensure that trucks remain safe to operate.

Drivers and Transport Staff

Truck Drivers

Responsible for transporting cargo and ensuring that deliveries are completed safely.

Logistics Assistants

Support drivers with loading and unloading activities.

Warehouse Operations

Warehouse Manager

Supervises warehouse activities including storage and cargo handling.

Warehouse Staff

Manage the loading, unloading, and organization of goods.

Finance and Administration

Finance Manager

Oversees financial operations including invoicing and payment processing.

Administrative Staff

Manage documentation and customer records.

IT and Database Management (New Department)

Database Administrator (DBA)

Manages the database system, ensures data security, and performs backups.

IT Support Specialist

Maintain the company's digital infrastructure.

3. Define Problems and Constraints

3.1 Company Problems

Faze Clan Movements currently faces several operational challenges due to its outdated record-keeping systems.

- **Manual Record Keeping**

Much of the company's operational information is recorded on paper or in isolated spreadsheets. This makes it difficult to retrieve information quickly and increases the likelihood of errors.

- **Poor Shipment Tracking**

Without a centralized system, employees and customers cannot easily track the real-time status of shipments.

- **Inefficient Fleet Management**

Maintenance schedules, inspections, and fuel consumption are not consistently monitored, which may lead to unexpected vehicle breakdowns and increased operating costs.

- **Data Redundancy and Inconsistency**

Because information is stored across multiple spreadsheets, duplicate records may exist and inconsistencies can occur.

- **Slow Reporting and Decision Making**

Finds it difficult to generate accurate reports on deliveries, driver performance, and operational costs.

- **Customer Communication Issues**

Employees may struggle to provide customers with accurate shipment updates due to fragmented information systems.

3.2 Company Constraints

Several constraints may affect the development and implementation of the database system.

- **Budget Constraints**

The company has limited financial resources available for purchasing software, hardware, and providing system maintenance.

- **Technical Constraints**

The database must be designed using Oracle SQL Developer as required for the academic project.

- **Time Constraints**

The design and implementation of the database must be completed within the time allocated for the project.

- **Training Constraints**

Employees may require training to effectively use the new database system.

- **Infrastructure Constraints**

The system must operate using the company's current computer hardware and network infrastructure.

4. Database System Specifications

4.1 Objectives to Solve Problems Identified

The primary objective of the proposed database system is to improve the efficiency and reliability of logistics operations at Faze Clan Movements.

The database system will achieve the following:

- The system will replace manual record keeping with a centralized digital database.
- It will support shipment tracking, driver and vehicle assignment, and fleet monitoring including maintenance, inspections, and fuel consumption.
- Enhance communication between departments and improve customer service.
- Assist with financial management by tracking invoices, payments, and transportation costs.
- Management will be able to generate reports that provide insights into operational performance, allowing them to make better strategic decisions.

4.2 Information Required from the Database

The database system must store and manage the following information:

Customer Information

- Customer names and contact details
- Customer addresses
- Customer shipment history

Shipment Information

- Shipment ID and cargo details
- Pickup and delivery locations
- Shipment status and delivery times

- Assigned driver, truck, and route

Cargo Information

- Cargo items being transported
- Cargo weight and description
- Cargo handling requirements

Driver Information

- Driver personal details
- Driver license information
- Driver availability and assigned deliveries

Vehicle Information

- Truck registration numbers
- Truck type and cargo capacity
- Vehicle maintenance and inspection records

Route Information

- Delivery routes and distances
- Estimated delivery times

Warehouse Information

- Warehouse locations
- Cargo storage records
- Shipment dispatch and arrival records

Fleet Maintenance Information

- Vehicle maintenance schedules
- Inspection records
- Fuel usage records

Dispatch and Scheduling Information

- Driver and truck assignments
- Delivery schedules
- Dispatch records

Financial Information

- Shipment costs and invoices
- Customer payments and outstanding balances

4.3 Scope

The scope of the Faze Clan Movements database system includes the following operational functions:

- Maintain customer records
- Maintain customer address records
- Maintain driver information
- Maintain employee records
- Maintain vehicle and fleet records
- Maintain vehicle maintenance and inspection records
- Maintain warehouse records
- Maintain shipment and cargo item records
- Maintain delivery records
- Maintain route information
- Track vehicle fuel usage
- Record dispatch schedules and truck assignments
- Generate invoices and manage customer payments
- Assign drivers and vehicles to shipments
- Generate operational reports including shipment performance, driver activity, and financial summaries

4.4 Boundaries

The following boundaries define the limits of the proposed database system:

Budget Boundary

The database system will be implemented within the budget allocated to the company for the entire project, including the budget for the software, hardware, and training.

Technology Boundary

The system must be developed using Oracle SQL Developer as required for the project.

Operational Boundary

The database system will be designed to cover transportation operations and the coordination of the warehouse only. Supplier manufacturing systems will be excluded from the system.

Time Boundary

The database system will be designed, developed, and implemented within the allocated time for the project.

Personnel Boundary

The system must be easy to use and require minimal technical expertise for daily operations.